

4.27) $x(t) = 3\cos(7\omega t) + 4$

a) medelvärde? medelvärde TEKNIS $\bar{x} = \frac{1}{T} \int_0^T f(\xi) d\xi$

$$\begin{aligned}\bar{x} &= \frac{1}{T} \int_0^T (3\cos(7\omega t) + 4) dt = \\ &= \frac{1}{T} \left[\frac{3}{7\omega} \sin(7\omega t) + 4t \right]_0^T = \frac{7\omega}{2\pi} \left[\frac{3}{7\omega} \sin(7\omega t) + 4t \right]_0^{\frac{2\pi}{7\omega}} = \\ &= \frac{7\omega}{2\pi} (0 + 4 \frac{2\pi}{7\omega} - 0 - 0) = \underline{\underline{4}}\end{aligned}$$

$\bar{x} = 4$

b) RMS-värde? RMS-värde TEKNIS $x_{rms} = \sqrt{\frac{1}{T} \int_0^T f^2(\xi) d\xi}$
VIFÄR

$$\begin{aligned}x_{rms}^2 &= \frac{1}{T} \int_0^T (3\cos(7\omega t) + 4)^2 dt = \\ &= \frac{1}{T} \int_0^T (9\cos^2(7\omega t) + 12\cos(7\omega t) + 16) dt = \\ &= \frac{7\omega}{2\pi} \int_0^{\frac{2\pi}{7\omega}} \left(\frac{9}{2}(1 + \cos(14\omega t)) + 12\cos(7\omega t) + 16 \right) dt = \\ &= \frac{7\omega}{2\pi} \left[\frac{9}{2}t + \frac{9}{2} \frac{1}{14\omega} \sin(14\omega t) + \frac{12}{7\omega} \sin(7\omega t) + 16t \right]_0^{\frac{2\pi}{7\omega}} = \\ &= \left(\frac{7\omega}{2\pi} \right)^2 \left(\frac{9}{2} + 16 \right) = \left(\frac{7\omega}{2\pi} \right)^2 41\end{aligned}$$

SVAR: a) $\bar{x} = 4$

b) $x_{rms} = \frac{7\omega}{2\pi} \sqrt{41}$